

Auteur: Noran Ben Said
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$$f(x) = \frac{x^2 - x - 2}{x^2 + 5x + 6}$$

$$f'(x) = \frac{(2x - 1)(x^2 + 5x + 6) - (x^2 - x - 2)(2x + 5)}{(x^2 + 5x + 6)^2}$$

$$(2x - 1)(x^2 + 5x + 6) = 2x^3 + 9x^2 + 7x - 6$$

$$(x^2 - x - 2)(2x + 5) = 2x^3 + 3x^2 - 9x - 10$$

$$(2x^3 + 9x^2 + 7x - 6) - (2x^3 + 3x^2 - 9x - 10)$$

$$= 6x^2 + 16x + 4$$

$$f'(x) = \frac{6x^2 + 16x + 4}{(x^2 + 5x + 6)^2}$$

References